

Certificate of Calibration

Report Number: 935503

Sensor Model: CX-1080-CU-HT-20L	Serial Number: X115888
Sensor Type: Cernox Resistor	Calibration Date: October 12, 2016
Sensor Excitation: see <i>Test Data</i> page of report	Calibration Due:
Temperature Range: 20.0 K to 325 K	

Traceability and Calibration Method

This temperature sensor has been calibrated to the International Temperature Scale of 1990 (ITS-90) or the Provisional Low Temperature Scale (PLTS-2000) as appropriate. The calibrations are traceable to the National Institute of Standards and Technology (NIST, United States), the National Physical Laboratory (NPL, United Kingdom), the Physikalisch-Technische Bundesanstalt (PTB, Germany), or natural physical constants.

Lake Shore Cryotronics maintains ITS-90 and PLTS-2000 on standard platinum (PRT), rhodium-iron (RIRT), and germanium (GRT) resistance thermometers that have been calibrated directly by an internationally recognized national metrology institute (NIST, NPL, PTB) for $T < 330$ K or an ISO 17025 accredited metrology laboratory for $330 \text{ K} < T < 800$ K. A nuclear orientation thermometer is also used for temperatures less than 50 mK. These standards are routinely intercompared to verify consistency and accuracy of the temperature scale.

The sensor calibrations are performed by comparison to laboratory standard resistance thermometers and tested in accordance with Lake Shore Cryotronics, Inc. Quality Assurance Manual (QP-4220). The quality system of Lake Shore Cryotronics is registered to ISO 9001.

Procedures used: 021-97-02, 099-00-00, 121-96-02, 029-95-02

Notes

The calibration results in this report apply only to the specific sensor specified above.

This report shall not be reproduced, except in full, without written approval from Lake Shore Cryotronics, Inc.

Unless stated otherwise, the uncertainties in this report are based on an approximate 95% confidence level with a coverage factor $k=2$.

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Calibration
Engineer/Technician

Approved by: Romerero Prince
Metrology



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F010-04-00_C

DATA PLOT

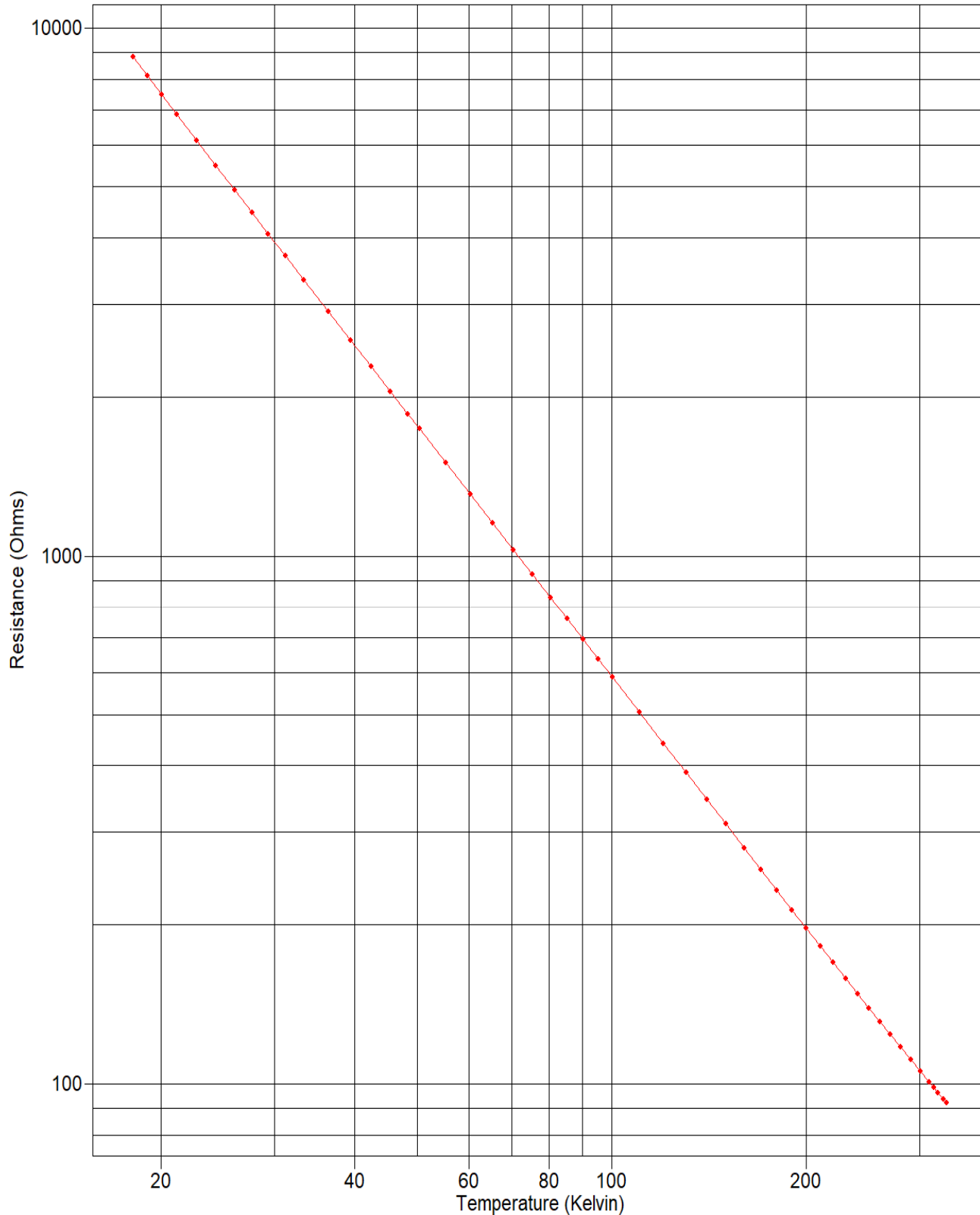
Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K



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TEST DATA

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Index	Temp. (K)	Resistance (Ω)	Excitation	Index	Temp. (K)	Resistance (Ω)	Excitation
1	18.1242	8818.99	2mV $\pm$ 25%	31	140.247	345.625	2mV $\pm$ 25%
2	19.1007	8097.35	2mV $\pm$ 25%	32	150.241	309.893	2mV $\pm$ 25%
3	20.0798	7468.22	2mV $\pm$ 25%	33	160.244	279.790	2mV $\pm$ 25%
4	21.1643	6862.02	2mV $\pm$ 25%	34	170.245	254.229	2mV $\pm$ 25%
5	22.7418	6115.54	2mV $\pm$ 25%	35	180.246	232.292	2mV $\pm$ 25%
6	24.3689	5477.11	2mV $\pm$ 25%	36	190.242	213.345	2mV $\pm$ 25%
7	26.0425	4927.99	2mV $\pm$ 25%	37	200.239	196.840	2mV $\pm$ 25%
8	27.7095	4466.64	2mV $\pm$ 25%	38	210.251	182.352	2mV $\pm$ 25%
9	29.3641	4074.95	2mV $\pm$ 25%	39	220.245	169.610	2mV $\pm$ 25%
10	31.2172	3698.97	2mV $\pm$ 25%	40	230.232	158.307	2mV $\pm$ 25%
11	33.3608	3330.84	2mV $\pm$ 25%	41	240.237	148.237	2mV $\pm$ 25%
12	36.3715	2905.93	2mV $\pm$ 25%	42	250.234	139.232	2mV $\pm$ 25%
13	39.3818	2563.40	2mV $\pm$ 25%	43	260.226	131.135	2mV $\pm$ 25%
14	42.3695	2284.43	2mV $\pm$ 25%	44	270.233	123.826	2mV $\pm$ 25%
15	45.3634	2051.44	2mV $\pm$ 25%	45	280.234	117.224	2mV $\pm$ 25%
16	48.3587	1854.81	2mV $\pm$ 25%	46	290.249	111.210	2mV $\pm$ 25%
17	50.3450	1740.92	2mV $\pm$ 25%	47	300.258	105.730	2mV $\pm$ 25%
18	55.3428	1499.72	2mV $\pm$ 25%	48	310.261	100.733	2mV $\pm$ 25%
19	60.3363	1308.94	2mV $\pm$ 25%	49	315.264	98.4003	2mV $\pm$ 25%
20	65.3288	1154.86	2mV $\pm$ 25%	50	320.259	96.1567	2mV $\pm$ 25%
21	70.3249	1028.34	2mV $\pm$ 25%	51	326.256	93.5937	2mV $\pm$ 25%
22	75.3162	923.024	2mV $\pm$ 25%	52	330.273	91.9419	2mV $\pm$ 25%
23	80.3040	834.373	2mV $\pm$ 25%				
24	85.2978	758.671	2mV $\pm$ 25%				
25	90.2831	693.605	2mV $\pm$ 25%				
26	95.2798	637.128	2mV $\pm$ 25%				
27	100.280	587.680	2mV $\pm$ 25%				
28	110.274	505.753	2mV $\pm$ 25%				
29	120.267	440.888	2mV $\pm$ 25%				
30	130.258	388.562	2mV $\pm$ 25%				

UNCERTAINTY ANALYSIS

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Calibration Data Uncertainty

The uncertainties of the measured calibration data for Lake Shore's sensors are summarized in the table below. The values given are the combined uncertainty of the temperature measurement and the resistance or voltage measurement expressed as an equivalent temperature uncertainty in millikelvin (mK). Note that the values are the calibration uncertainty only and do not include the stability of the temperature sensor. The uncertainty analysis has followed the guidelines for determining measurement uncertainty as outlined in the ISO Guide to the Expression of Uncertainty in Measurement, NIST Technical Note 1297, and ANSI/NCSS Z540-2-1997. Since the uncertainty varies with temperature due to the variation of the sensor sensitivity and excitation, the table gives typical values at several different temperatures throughout the range of the calibration. The uncertainty is based on an approximate 95% confidence level with a coverage factor $k = 2$.

T (K)	Uncertainty ($\pm$ mK)												
	GR	Cernox (CX)					RX			Platinum		RF-800	Diode
		1010	1030	1050	1070	1080	102A	103A	202A	100 Ω	25 Ω	27 Ω	
1.4	4	4	4	4			4	4	4			5	7
4.2	4	4	4	4	4		4	6	5			5	5
10	4	5	5	4	4		10	15	12			7	6
20	8	10	9	8	8	8	35	35	28	9	10	13	9
30	9	13	11	9	9	9	76	61	46	9	9	14	31
50	11	18	14	12	12	11				10	10	13	37
100	20	29	22	17	16	14				11	12	12	32
300		78	60	46	45	36				24	24	25	35
400		124	94	74	72	60				45	45	45	49
500										51	51		54

Polynomial Fit Uncertainty

When a sensor is used to measure temperature, a polynomial fit to the measured calibration data is often used to convert the sensor resistance (R) or voltage (V) to a temperature (T). How well the polynomial represents the sensor calibration data is another source of uncertainty when using the sensor. In the polynomials provided with this set of calibration data, the standard deviation of the fit can be used as an estimate of this additional temperature uncertainty. The standard deviation of fit is determined from the following equation:

$$\sigma_{fit}^2 = \frac{\sum_{i=1}^N (T_i - T_{i,calc})^2}{N - n} = \frac{N}{N - n} (\Delta T_{RMS})^2$$

where

σ_{fit} = standard deviation of the fit

T_i = measured temperature for point i

$T_{i,calc}$ = the temperature calculated from the polynomial equation for point i

N = number of data points in fit range

n = number of fit coefficients

ΔT_{RMS} = root mean square deviation of fit

A value of ΔT_{RMS} is given for each range of fit.

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POLYNOMIAL EQUATION

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

20.0 K to 95.3 K
7516 ohms to 637.1 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 2.70393846939

ZU = 3.94541884748

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	54.206273	2.7951E-04	193934.85
1	-44.634649	4.5132E-04	-98897.42
2	9.832668	4.1992E-04	23415.71
3	-1.430067	3.9568E-04	-3614.23
4	0.160306	3.7236E-04	430.51
5	-0.009964	3.6073E-04	-27.62
6	-0.000076	3.4272E-04	-0.22

$Z = \text{Log}(\text{Resistance})$

$k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)$

Temp. (K) = $\sum A_i * \text{COS}(i * \text{ARCCOS}(k))$, where $0 \leq i \leq 6$
and the A_i 's are the coefficients in the table above.

POLYNOMIAL EQUATION

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Temp. (K) vs. Log(Resistance)

	R Meas. (W)	T Meas. (K)	T Eq. (K)	T diff. (mK)
1	8818.990	18.12418	18.12449	-0.31
2	8097.350	19.10070	19.10051	0.19
3	7468.218	20.07981	20.07964	0.17
4	6862.023	21.16425	21.16383	0.42
5	6115.540	22.74181	22.74152	0.29
6	5477.115	24.36895	24.36939	-0.44
7	4927.989	26.04249	26.04406	-1.58
8	4466.637	27.70951	27.70940	0.11
9	4074.953	29.36408	29.36316	0.92
10	3698.969	31.21715	31.21706	0.09
11	3330.836	33.36078	33.35919	1.59
12	2905.929	36.37146	36.37240	-0.94
13	2563.397	39.38183	39.38320	-1.36
14	2284.428	42.36947	42.36960	-0.12
15	2051.440	45.36344	45.36286	0.57
16	1854.808	48.35872	48.35872	0.00
17	1740.923	50.34500	50.34354	1.46
18	1499.724	55.34281	55.34298	-0.17
19	1308.944	60.33631	60.33656	-0.25
20	1154.859	65.32875	65.33047	-1.72
21	1028.341	70.32493	70.32483	0.09
22	923.0238	75.31619	75.31728	-1.08
23	834.3731	80.30399	80.30145	2.54
24	758.6708	85.29780	85.29573	2.07
25	693.6046	90.28306	90.28585	-2.79
26	637.1284	95.27982	95.27868	1.14
27	587.6796	100.27951	100.28095	-1.44
28	505.7530	110.27441	110.27385	0.56

Order of Fit = 6 RMS error of fit = 1.17 mK

Largest absolute error = -2.79 mK at data point no. 25

POLYNOMIAL EQUATION

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Useful Range of Fit:

95.3 K to 325 K
637.1 ohms to 94.12 ohms

Lower and Upper limits of Log(Resistance) used in computing Chebychev coefficients:

ZL = 1.96351352283

ZU = 2.88005339749

Order	Coefficient	Std. Deviation of Coefficient	Ratio (Coeff./Std Dev.)
0	186.486973	9.6483E-04	193284.06
1	-119.394486	1.4714E-03	-81143.84
2	20.783852	1.3947E-03	14902.37
3	-3.006345	1.3874E-03	-2166.83
4	0.502049	1.3450E-03	373.27
5	-0.087095	1.2806E-03	-68.01
6	0.013914	1.2651E-03	11.00
7	-0.003151	1.2557E-03	-2.51

$Z = \text{Log}(\text{Resistance})$

$k = ((Z-ZL)-(ZU-Z))/(ZU-ZL)$

Temp. (K) = $\sum A_i * \text{COS}(i * \text{ARCCOS}(k))$, where $0 \leq i \leq 7$
and the A_i 's are the coefficients in the table above.

POLYNOMIAL EQUATION

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Polynomial Type: Chebychev

Temp. (K) vs. Log(Resistance)

	R Meas. (W)	T Meas. (K)	T Eq. (K)	T diff. (mK)
24	758.6708	85.29573	85.29571	0.02
25	693.6046	90.28585	90.28629	-0.44
26	637.1284	95.27868	95.27797	0.72
27	587.6796	100.27951	100.27948	0.03
28	505.7530	110.27441	110.27421	0.20
29	440.8879	120.26742	120.26947	-2.05
30	388.5624	130.25752	130.25698	0.54
31	345.6249	140.24733	140.24545	1.88
32	309.8935	150.24097	150.23946	1.51
33	279.7902	160.24430	160.24680	-2.50
34	254.2290	170.24470	170.24417	0.53
35	232.2921	180.24584	180.24784	-2.00
36	213.3453	190.24178	190.24137	0.41
37	196.8396	200.23861	200.23857	0.04
38	182.3522	210.25115	210.25113	0.02
39	169.6104	220.24505	220.24014	4.91
40	158.3068	230.23202	230.23724	-5.22
41	148.2365	240.23722	240.23643	0.80
42	139.2316	250.23412	250.22952	4.60
43	131.1353	260.22563	260.22932	-3.68
44	123.8264	270.23264	270.23858	-5.93
45	117.2241	280.23432	280.22717	7.15
46	111.2100	290.24871	290.24521	3.49
47	105.7300	300.25837	300.26605	-7.68
48	100.7325	310.26053	310.26720	-6.67
49	98.40032	315.26441	315.25299	11.41
50	96.15666	320.25855	320.25997	-1.43
51	93.59368	326.25556	326.25165	3.91
52	91.94191	330.27330	330.27787	-4.57

Order of Fit = 7 RMS error of fit = 4.06 mK

Largest absolute error = 11.41 mK at data point no. 49

INTERPOLATION TABLE

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>	<u>Temp (K)</u>	<u>Res. (Ω)</u>	<u>dR/dT (Ω/K)</u>	<u>dlogR/dlogT</u>
20.00	7516.27	-606.66	-1.6143	130.0	389.780	-4.7504	-1.5844
21.00	6948.24	-531.76	-1.6072	135.0	367.152	-4.3104	-1.5849
22.00	6448.57	-469.40	-1.6014	140.0	346.586	-3.9244	-1.5852
23.00	6006.13	-416.99	-1.5968	145.0	327.832	-3.5843	-1.5853
24.00	5611.97	-372.51	-1.5931	150.0	310.678	-3.2832	-1.5852
25.00	5258.95	-334.49	-1.5901	155.0	294.944	-3.0156	-1.5848
26.00	4941.25	-301.72	-1.5876	160.0	280.474	-2.7768	-1.5841
27.00	4654.07	-273.31	-1.5856	165.0	267.134	-2.5631	-1.5831
28.00	4393.42	-248.53	-1.5839	170.0	254.807	-2.3711	-1.5819
29.00	4155.99	-226.79	-1.5825	175.0	243.391	-2.1982	-1.5805
30.00	3938.97	-207.64	-1.5814	180.0	232.797	-2.0419	-1.5788
31.00	3739.98	-190.67	-1.5805	185.0	222.947	-1.9003	-1.5769
32.00	3556.99	-175.59	-1.5797	190.0	213.772	-1.7718	-1.5747
33.00	3388.26	-162.13	-1.5790	195.0	205.211	-1.6547	-1.5724
34.00	3232.27	-150.07	-1.5786	200.0	197.208	-1.5478	-1.5698
35.00	3087.72	-139.21	-1.5780	205.0	189.717	-1.4501	-1.5670
36.00	2953.49	-129.42	-1.5776	210.0	182.693	-1.3606	-1.5640
37.00	2828.56	-120.58	-1.5773	215.0	176.099	-1.2784	-1.5608
38.00	2712.06	-112.55	-1.5770	220.0	169.899	-1.2027	-1.5574
39.00	2603.22	-105.24	-1.5767	225.0	164.062	-1.1330	-1.5538
40.00	2501.36	-98.587	-1.5765	230.0	158.560	-1.0686	-1.5501
42.00	2316.19	-86.918	-1.5761	235.0	153.368	-1.0091	-1.5462
44.00	2152.45	-77.090	-1.5758	240.0	148.462	-0.95395	-1.5421
46.00	2006.85	-68.741	-1.5757	245.0	143.821	-0.90280	-1.5379
48.00	1876.69	-61.595	-1.5754	250.0	139.428	-0.85529	-1.5336
50.00	1759.80	-55.445	-1.5753	255.0	135.263	-0.81109	-1.5291
52.00	1654.36	-50.116	-1.5752	260.0	131.312	-0.76992	-1.5245
54.00	1558.89	-45.468	-1.5750	265.0	127.559	-0.73152	-1.5197
56.00	1472.10	-41.404	-1.5750	270.0	123.992	-0.69567	-1.5149
58.00	1392.95	-37.825	-1.5750	273.15	121.835	-0.67429	-1.5117
60.00	1320.53	-34.662	-1.5749	275.0	120.599	-0.66214	-1.5099
65.00	1164.12	-28.207	-1.5750	280.0	117.367	-0.63076	-1.5048
70.00	1035.87	-23.310	-1.5752	285.0	114.288	-0.60136	-1.4996
75.00	929.183	-19.519	-1.5755	290.0	111.351	-0.57378	-1.4943
77.35	885.093	-18.031	-1.5758	295.0	108.547	-0.54789	-1.4890
80.00	839.334	-16.535	-1.5760	300.0	105.869	-0.52355	-1.4836
85.00	762.837	-14.149	-1.5766	305.0	103.309	-0.50065	-1.4781
90.00	697.083	-12.218	-1.5775	310.0	100.860	-0.47909	-1.4725
95.00	640.074	-10.636	-1.5786	315.0	98.5163	-0.45877	-1.4669
100.0	590.276	-9.3232	-1.5795	320.0	96.2708	-0.43960	-1.4612
105.0	546.486	-8.2241	-1.5802	325.0	94.1185	-0.42150	-1.4555
110.0	507.748	-7.2975	-1.5810				
115.0	473.279	-6.5104	-1.5819				
120.0	442.456	-5.8360	-1.5828				
125.0	414.764	-5.2548	-1.5837				

THERMAL CYCLE TESTING

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Serial Number: X115888

Sensor Type: Cernox Resistor

This sensor was tested for repeatability through rapid thermal cycles from room temperature into liquid helium. During this test, the following four lead resistance values were recorded:

Approximately 295 K:	108 Ω
Liquid Nitrogen:	887 Ω
Liquid Helium:	175667 Ω

The nitrogen and helium values were recorded in OPEN dewars, so precision comparisons with calibration values or other thermal cycle test values should not be made.

Recommended Operating Parameters:

For sensors calibrated by Lake Shore, the current to the sensor is adjusted to maintain the sensor output voltage or power at the values listed on the Test Data page.

BREAKPOINTS CUBIC SPLINE FORMAT

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Sensor Model: CX-1080-CU-HT-20L
Serial Number: X115888
Data Format: 7 (Ohms/Kelvin)
Setpoint Limit: 325

Measurement (ohms)	Temp (K)	Curvature
9.19419E+01	3.30278E+02	5.02610E-02
9.35937E+01	3.26252E+02	4.77876E-02
9.61567E+01	3.20260E+02	4.39497E-02
9.84003E+01	3.15253E+02	4.10105E-02
1.00733E+02	3.10267E+02	3.81375E-02
1.05730E+02	3.00266E+02	3.29320E-02
1.11210E+02	2.90245E+02	2.83153E-02
1.17224E+02	2.80227E+02	2.42144E-02
1.23826E+02	2.70239E+02	2.06052E-02
1.31135E+02	2.60229E+02	1.74257E-02
1.39232E+02	2.50230E+02	1.46485E-02
1.48237E+02	2.40236E+02	1.22337E-02
1.58307E+02	2.30237E+02	1.01432E-02
1.69610E+02	2.20240E+02	8.34589E-03
1.82352E+02	2.10251E+02	6.81079E-03
1.96840E+02	2.00239E+02	5.50480E-03
2.13345E+02	1.90241E+02	4.40640E-03
2.32292E+02	1.80248E+02	3.48859E-03
2.54229E+02	1.70244E+02	2.72771E-03
2.79790E+02	1.60247E+02	2.10418E-03
3.09893E+02	1.50239E+02	1.59813E-03
3.45625E+02	1.40245E+02	1.19352E-03
3.88562E+02	1.30257E+02	8.74055E-04
4.40888E+02	1.20269E+02	6.25969E-04
5.05753E+02	1.10274E+02	4.35176E-04
5.87680E+02	1.00279E+02	2.95599E-04
6.37128E+02	9.52787E+01	2.40043E-04
6.93605E+02	9.02858E+01	1.92636E-04
7.58671E+02	8.52957E+01	1.52088E-04
8.34373E+02	8.03014E+01	1.18461E-04
9.23024E+02	7.53173E+01	9.07878E-05
1.02834E+03	7.03248E+01	6.82708E-05
1.15486E+03	6.53305E+01	5.02393E-05
1.30894E+03	6.03366E+01	3.60741E-05
1.49972E+03	5.53430E+01	2.51129E-05
1.74092E+03	5.03435E+01	1.70124E-05
1.85481E+03	4.83587E+01	1.45513E-05
2.05144E+03	4.53629E+01	1.10842E-05
2.28443E+03	4.23696E+01	8.35385E-06
2.56340E+03	3.93832E+01	6.15469E-06
2.90593E+03	3.63724E+01	4.41142E-06
3.33084E+03	3.33592E+01	3.08535E-06
3.69897E+03	3.12171E+01	2.35600E-06
4.07495E+03	2.93632E+01	1.82604E-06
4.46664E+03	2.77094E+01	1.43539E-06
4.92799E+03	2.60441E+01	1.10710E-06
5.47711E+03	2.43694E+01	8.38366E-07
6.11554E+03	2.27415E+01	6.26776E-07
6.86202E+03	2.11638E+01	4.64384E-07
7.46822E+03	2.00796E+01	3.72277E-07
8.09735E+03	1.91005E+01	3.05067E-07
8.81899E+03	1.81245E+01	2.27975E-07



BREAKPOINTS 340 FORMAT

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Name: CX-1080-CU-HT-20L

Serial Number: X115888

Format: 4 ;Log Ohms/Kelvin

Limit: 325.0

Coefficient: 1 ;Negative

Point 1: 1.97365,325.000	Point 51: 2.80260, 95.500
Point 2: 1.98546,319.000	Point 52: 2.82078, 93.000
Point 3: 1.99653,313.500	Point 53: 2.83945, 90.500
Point 4: 2.00784,308.000	Point 54: 2.85864, 88.000
Point 5: 2.01941,302.500	Point 55: 2.87838, 85.500

Point 6: 2.03123,297.000	Point 56: 2.89869, 83.000
Point 7: 2.04333,291.500	Point 57: 2.91541, 81.000
Point 8: 2.05570,286.000	Point 58: 2.93252, 79.000
Point 9: 2.06836,280.500	Point 59: 2.95006, 77.000
Point 10: 2.08132,275.000	Point 60: 2.96807, 75.000

Point 11: 2.09459,269.500	Point 61: 2.98656, 73.000
Point 12: 2.10694,264.500	Point 62: 3.00557, 71.000
Point 13: 2.11956,259.500	Point 63: 3.02511, 69.000
Point 14: 2.13246,254.500	Point 64: 3.04523, 67.000
Point 15: 2.14566,249.500	Point 65: 3.06596, 65.000

Point 16: 2.15916,244.500	Point 66: 3.08733, 63.000
Point 17: 2.17299,239.500	Point 67: 3.10940, 61.000
Point 18: 2.18714,234.500	Point 68: 3.13219, 59.000
Point 19: 2.20163,229.500	Point 69: 3.15339, 57.200
Point 20: 2.21648,224.500	Point 70: 3.17525, 55.400

Point 21: 2.23170,219.500	Point 71: 3.19784, 53.600
Point 22: 2.24730,214.500	Point 72: 3.22121, 51.800
Point 23: 2.26331,209.500	Point 73: 3.24540, 50.000
Point 24: 2.27974,204.500	Point 74: 3.27048, 48.200
Point 25: 2.29660,199.500	Point 75: 3.29359, 46.600

Point 26: 2.31217,195.000	Point 76: 3.31749, 45.000
Point 27: 2.32812,190.500	Point 77: 3.34226, 43.400
Point 28: 2.34448,186.000	Point 78: 3.36797, 41.800
Point 29: 2.36126,181.500	Point 79: 3.39468, 40.200
Point 30: 2.37847,177.000	Point 80: 3.42072, 38.700

Point 31: 2.39615,172.500	Point 81: 3.44779, 37.200
Point 32: 2.41431,168.000	Point 82: 3.47598, 35.700
Point 33: 2.43297,163.500	Point 83: 3.50340, 34.300
Point 34: 2.45217,159.000	Point 84: 3.53196, 32.900
Point 35: 2.47192,154.500	Point 85: 3.56178, 31.500

Point 36: 2.49227,150.000	Point 86: 3.59073, 30.200
Point 37: 2.51089,146.000	Point 87: 3.62094, 28.900
Point 38: 2.53000,142.000	Point 88: 3.65259, 27.600
Point 39: 2.54968,138.000	Point 89: 3.68321, 26.400
Point 40: 2.56992,134.000	Point 90: 3.71527, 25.200

Point 41: 2.59077,130.000	Point 91: 3.74898, 24.000
Point 42: 2.61227,126.000	Point 92: 3.78150, 22.900
Point 43: 2.63445,122.000	Point 93: 3.81567, 21.800
Point 44: 2.65737,118.000	Point 94: 3.85177, 20.700
Point 45: 2.68106,114.000	Point 95: 3.87597, 20.000

Point 46: 2.70249,110.500
Point 47: 2.72458,107.000
Point 48: 2.74740,103.500
Point 49: 2.76760,100.500
Point 50: 2.78489, 98.000



BREAKPOINTS 91C/93C/330 FORMAT

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Interpolation Method: Lagrangian

Limit: 325.0 (Kelvin)

Format: 4 (Log Ohms/Kelvin)

Number of Breakpoints: 35

No.	Units	Temperature (K)	No.	Units	Temperature (K)
1	1.97368	325.0	21	3.18651	54.5
2	1.97562	324.0	22	3.25234	49.5
3	2.00579	309.0	23	3.29365	46.6
4	2.03781	294.0	24	3.34549	43.2
5	2.07189	279.0	25	3.39306	40.3
6	2.10821	264.0	26	3.44237	37.5
7	2.14702	249.0	27	3.49160	34.9
8	2.18860	234.0	28	3.54045	32.5
9	2.23327	219.0	29	3.58855	30.3
10	2.28143	204.0	30	3.63791	28.2
11	2.33356	189.0	31	3.68593	26.3
12	2.39024	174.0	32	3.73486	24.5
13	2.45221	159.0	33	3.78162	22.9
14	2.52042	144.0	34	3.82871	21.4
15	2.59613	129.0	35	3.87600	20.0
16	2.68112	114.0			
17	2.77795	99.0			
18	2.89053	84.0			
19	3.02515	69.0			
20	3.12647	59.5			

Temperature for Resistance Decades:

Res. (Ohms)	Temp. (K)
100	311.805
1000	71.578

BREAKPOINTS 234 FORMAT

Calibration Report: 935503

Sensor Model: CX-1080-CU-HT-20L

Sensor Type: Cernox Resistor

Serial Number: X115888

Temperature Range: 20.0 K to 325 K

Maximum Temperature Error:

1.4 - 10 K: -
10 - 20 K: 0.005 K
20 - 40 K: 0.011 K
40 - 100 K: 0.023 K
> 100 K: 0.029 K

BP #	Temp. (K)	Res. (W)	Log10 Res.	BP #	Temp. (K)	Res. (W)	Log10 Res.
1	321.768	95.49926	1.980	41	98.704	602.5596	2.780
2	311.810	100.0000	2.000	42	95.867	630.9573	2.800
3	302.231	104.7129	2.020	43	93.111	660.6934	2.820
4	293.009	109.6478	2.040	44	90.432	691.8310	2.840
5	284.126	114.8154	2.060	45	87.830	724.4360	2.860
6	275.564	120.2264	2.080	46	85.302	758.5776	2.880
7	267.305	125.8925	2.100	47	82.847	794.3282	2.900
8	259.335	131.8257	2.120	48	80.461	831.7638	2.920
9	251.639	138.0384	2.140	49	78.144	870.9636	2.940
10	244.203	144.5440	2.160	50	75.893	912.0108	2.960
11	237.016	151.3561	2.180	51	73.707	954.9926	2.980
12	230.066	158.4893	2.200	52	71.584	1000.000	3.000
13	223.342	165.9587	2.220	53	67.518	1096.478	3.040
14	216.834	173.7801	2.240	54	63.683	1202.264	3.080
15	210.533	181.9701	2.260	55	60.066	1318.257	3.120
16	204.430	190.5461	2.280	56	56.654	1445.440	3.160
17	198.517	199.5262	2.300	57	53.436	1584.893	3.200
18	192.786	208.9296	2.320	58	50.401	1737.801	3.240
19	187.230	218.7762	2.340	59	47.539	1905.461	3.280
20	181.842	229.0868	2.360	60	44.839	2089.296	3.320
21	176.615	239.8833	2.380	61	42.294	2290.868	3.360
22	171.544	251.1886	2.400	62	39.894	2511.886	3.400
23	166.623	263.0268	2.420	63	37.630	2754.229	3.440
24	161.846	275.4229	2.440	64	35.496	3019.952	3.480
25	157.209	288.4032	2.460	65	33.484	3311.311	3.520
26	152.707	301.9952	2.480	66	31.587	3630.781	3.560
27	148.334	316.2278	2.500	67	29.799	3981.072	3.600
28	144.087	331.1311	2.520	68	28.114	4365.158	3.640
29	139.962	346.7369	2.540	69	26.527	4786.301	3.680
30	135.954	363.0781	2.560	70	25.033	5248.075	3.720
31	132.060	380.1894	2.580	71	23.626	5754.399	3.760
32	128.277	398.1072	2.600	72	22.302	6309.573	3.800
33	124.601	416.8694	2.620	73	21.056	6918.310	3.840
34	121.029	436.5158	2.640	74	19.886	7585.776	3.880
35	117.558	457.0882	2.660	75	18.787	8317.638	3.920
36	114.186	478.6301	2.680				
37	110.909	501.1872	2.700				
38	107.724	524.8075	2.720				
39	104.630	549.5409	2.740				
40	101.625	575.4399	2.760				